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SHORT COMMUNICATION

Functional ecology of growth in seedlings versus root sprouts of *Fagus grandifolia* Ehrh.

Emad Farahat · Martin J. Lechowicz

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Abstract We assessed growth differences and the foliar traits underpinning production in paired samples of juvenile American beech (*Fagus grandifolia* Ehrh.) that originated from seed versus root sprouts. Root sprouts had significantly greater relative extension growth rate and slightly greater leaf mass per unit area compared to seed-derived individuals, but neither light-saturated net photosynthetic rate nor foliar chlorophyll and nitrogen concentrations differed significantly between paired seedlings and sprouts. The greater height growth rate of saplings originating as root sprouts does not result from differing foliar function, but rather depends on translocation of assimilates from the parent tree to sustain this unusual and ecologically important dual regeneration strategy in American beech.

Keywords *Fagus grandifolia* · Root sprouts · Seedlings · Leaf mass per area (LMA) · Chlorophyll content · Foliar nitrogen · Light-saturated net photosynthetic rate

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E. Farahat (✉) · M. J. Lechowicz
Department of Biology, McGill University, 1205 Avenue Dr. Penfield, H3A1B1 Montréal, QC, Canada
e-mail: emad23_1999@yahoo.com

M. J. Lechowicz
e-mail: martin.lechowicz@mcgill.ca

Present Address:
E. Farahat
Department of Botany and Microbiology, Faculty of Science, Helwan University, Cairo, Egypt

Introduction

Beech species (*Fagus* L., Fagaceae) are among the most representative trees in the temperate deciduous forests of the Northern Hemisphere (Wagner et al. 2010). American beech (*Fagus grandifolia* Ehrh.), a canopy dominant of the pre-settlement forest in many parts of eastern North America (Tubbs and Houston 1990), has the widest thermal range of all the beech species (Fang and Lechowicz 2006). American beech also is the only *Fagus* species that under natural conditions reproduces by sprouts from the lateral roots of a parent tree (Wagner et al. 2010). Larger parent trees have more root sprouts (Jones and Raynal 1987, 1988). Since seed production and seedling establishment in American beech varies considerably among sites and years (Dix and Skrentny 1965; Forcier 1973; Cleavitt et al. 2008; Wagner et al. 2010), successful regeneration often depends on establishment and survival of both seedlings and root sprouts (Takahashi and Lechowicz 2008; Takahashi et al. 2010). The proportion of root sprouts versus seedlings at a site varies, sprouting generally being more common at sites northward in the range (Kitamura and Kawano 2001) and under more disturbed or harsh environmental conditions (Held 1983; Morris et al. 2004). At 20 sites near the northern limit of the range, 58–91 % of the juvenile American beech were of sprout origin (Nyland 2008), and at another northern site, American beech mostly reproduced by sprouts along a slope gradient except under the more favorable conditions on the lower slope (Takahashi et al. 2010). While it is clear that root sprouting is an essential element of regeneration in American beech (Takahashi et al. 2010), the functional relationship between sprouts and their parent tree is less certain.

Translocation between saplings derived from root sprouts and their parent tree is documented (Kochenderfer et al. 2004, 2006) and such translocation can be critical to growth and survival (Wiehle et al. 2009). Saplings originating as sprouts as opposed to seedlings (Ward 1961; Houston 2001; Beaudet et al. 2007; Beaudet and Messier 2008) often, but not always (Canham 1988; Takahashi and Lechowicz 2008), have greater growth rates than seedlings. While a greater growth rate by sprouts compared to seedlings is consistent with an inference of parental support to the net advantage of sprouts, the possibility exists that saplings of sprout origin sometimes may function as a shade-adapted subcanopy contributing to the growth and maintenance of the parent tree. If so, we might expect some differentiation between sprouts versus seedlings in foliar characteristics associated with growth and production, but the few available studies offer no evidence of clear functional differences (Beaudet and Messier 2008; Takahashi and Lechowicz 2008). Previous studies, however, have compared seedlings and sprouts at the population level or even across sites, thus confounding the seedling–sprout contrast with possible microenvironmental effects on differences among sampled individuals. Our objectives in this study therefore were: (1) to document the degree to which immediately adjacent saplings derived from seed versus root sprouting at a northern site differed in their growth in the forest understory and (2) to assess whether differences in foliar traits between sprouts and seedlings might contribute to the observed growth differences.

Materials and methods

Study site and selection of paired samples

We compared 26 pairs of adjacent saplings of American beech originating as seedlings versus root sprouts and growing in the Gault Nature Reserve on Mont St. Hilaire (45°32′N, 73°8′W), the largest remaining tract of old-growth deciduous forest in southern Québec, Canada. Sugar maple (*Acer saccharum* Marshall) and American beech are codominant canopy trees throughout the reserve (Arii et al. 2005). We selected pairs of individuals as samples taking care that (1) sprout and seedling heights were near equal and no more than 125 cm and (2) the paired individuals were within 3 m of one another and in comparable microenvironments. Sprouts are differentiated from seedlings by emergence of the stem from a lateral root of the parent tree running in the upper soil horizon (Jones and Raynal 1987); we distinguished sprouts and seedlings in this study on that basis, by digging around each individual plant and removing the superficial soil temporarily to confirm the contact of root sprout to its parent tree. Based on the analysis of hemispheric canopy photos (WinsCANOPY, Régent Instruments, Ste-Foy, QC, Canada), the 26 sample sites represented the range of understory conditions in this forest with a gap fraction from 0.02 to 0.07 (median 0.06), 13–107 min of sunfleck exposure per day (median 57 min); R:FR ratios measured with a Skye SKR 110 R:FR sensor (Skye Instruments Ltd, UK) ranged from 0.07 to 0.27 (median 0.11) during the period of measurements.

Gas exchange measurements

We sampled photosynthetic light response curves of beech seedlings and sprouts in July and August 2007 on a fully mature leaf at the top, middle and bottom of each individual using a portable gas exchange system (Li-Cor 6400; Li-Cor, Inc., Lincoln, NE, USA). Leaf temperature (T °C) and relative humidity (RH %) of the leaf chamber were set automatically for T and manually controlled for RH to match ambient conditions. Cuvette CO_2 concentration was 400 μmol CO_2 mol^{-1}. Photon flux density (Q) was changed gradually in the sequence 0, 50, 100, 250, 500, 750, 1,000 and 1,500 μmol m^{-2} s^{-1}. We used Photosyn Assistant v. 1.2 software (Dundee Scientific, Dundee, Scotland, UK) to calculate the light-saturated net photosynthetic rate (A_{max}, μmol CO_2 m^{-2} s^{-1}) and dark respiration rate (R, μmol CO_2 m^{-2} s^{-1}).

Foliar characteristics

Foliar chlorophyll content (chl, g m^{-2}) was estimated on leaves at the time of photosynthetic assay with a chlorophyll meter (SPAD-502, Minolta Co., Osaka, Japan) as an average for all the intact leaves on each seedling or sprout. We used regression equations (Uemura et al. 2005) for both *F. crenata* and *F. japonica* to convert the SPAD readings to a chlorophyll estimate and took the average of the two estimates: Eq. 1 for *F. crenata* ($r^2 = 0.95$, $n = 92$)

$$\mathrm{Chl} = \exp\left(\frac{\mathrm{SPAD} - 54.61}{19.79}\right) - 0.05; \tag{1}$$

Equation 2 for *F. japonica* ($r^2 = 0.89$, $n = 88$).

$$\mathrm{Chl} = \exp\left(\frac{\mathrm{SPAD} - 52.33}{21.53}\right) - 0.08. \tag{2}$$

We harvested the three leaves of each seedling and sprout used for the photosynthetic assays to estimate leaf mass per area (LMA). We scanned each leaf and used ImagJ 1.38x (website: http://rsb.info.nih.gov/ij/java 1.6.0_02) to estimate fresh leaf area and measured leaf mass after drying at 60 °C. We analyzed foliar N concentrations of the harvested leaves using a Fisons EA 1108 CHNS-O Elemental Analyzer (Fisons, Milan, Italy).

Extension growth rate

Using terminal bud scars examined in August 2007 when extension growth had ended for the current season, we calculated the mean relative extension growth rate (South 1995) over the last three growing seasons as [ln (2007) length – ln (2005) length]/3.

Equivalence of moisture regime in the sprout–seedling pairs

We used a Dynamax Theta probe (Delta-T Devices, Cambridge) to measure the volumetric water content (Gaskin and Miller 1996) at the base of each beech sprout and seedling five times through the growing season, representing all possible soil moisture variations: saturated soil (rainy days), soil field capacity (1 day after rain) and low soil moisture content (3 days after rain and during the sampling measurements). Measurements were taken pairwise and each cycle of moisture estimates across all pairs was completed within 1.5 h. Neither these measurements nor visual inspection of the sampling sites suggested any difference in moisture regimes between the paired individuals.

Statistical analysis

We used the Wilcoxon matched pairs test to differentiate between the paired seedlings and root sprouts of American beech trees (Statistica 7.0 software, Statsoft Inc., USA).

Results and discussion

The extension growth of saplings originating as sprouts was twice that of nearby saplings originating as seedlings (Table 1). Because of the close proximity of the paired sprouts and saplings, this growth difference cannot be accounted for by microenvironmental variation. That leaves two possibilities: (1) parental subsidy of sprout growth or (2) differences in foliar function affecting the carbon and energy available for growth in sprouts versus seedlings. It is the case that the LMA of sprouts is greater than that of nearby seedlings which might suggest a degree of adaptation for more efficient use of sunflecks (Niinemets 2001). It seems unlikely that this slight difference in LMA will have a strong effect on the net production of adjacent seedlings and sprouts. Furthermore, other defining characteristics of foliar function such as light-saturated net photosynthetic rate, foliar chlorophyll content and the concentration of foliar nitrogen do not differ between adjacent seedlings and sprouts (Table 1), and the photosynthetic light response curves of the two types of saplings are identical (not shown). Hence, we conclude that foliar function in the leaves of root sprouts and seedlings in American beech is essentially the same and that the greater growth rate of sprouts therefore is driven by parental support. This parental support is consistent with documented functional traits of the root sprouting habit in terms of the persistence of American beech in competition with co-dominant *Acer saccharum* in northern forests (Takahashi and Lechowicz 2008; Takahashi et al. 2010).

Table 1 Mean ± standard deviation and Wilcoxon test for differences in paired seedlings and root sprouts of American beech trees

Trait	Mean ± SD		p (H_0: no difference)
	Seedling	Sprout	
RGR_{05-07} (cm cm^{-1} $year^{-1}$)	0.15 ± 0.06	0.33 ± 0.15	**<0.001**
LMA (g m^{-2})	23.3 ± 2.7	24.8 ± 2.8	**0.002**
A_{max} (μmol CO_2 m^{-2} s^{-1})	2.93 ± 0.59	3.08 ± 0.68	0.31
R (μmol CO_2 m^{-2} s^{-1})	0.13 ± 0.27	0.05 ± 0.35	0.36
Foliar chlorophyll (g m^2)	0.28 ± 0.05	0.27 ± 0.04	0.89
Foliar N (%)	1.78 ± 0.20	1.76 ± 0.28	0.80

A_{max} Light-saturated net photosynthetic rate, R dark respiration rate, *LMA* leaf mass per area, RGR_{05-07} the average relative extension growth rate over three growing seasons (2005–2007)

Bold numbers are significant at $p < 0.001$

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References

Arii K, Hamel BR, Lechowicz MJ (2005) Environmental correlates of canopy composition at Mont St. Hilaire, Quebec, Canada. J Torrey Bot Soc 132:90–102

Beaudet M, Messier C (2008) Beech regeneration of seed and root sucker origin: a comparison of morphology, growth, survival, and response to defoliation. For Ecol Manag 255:3659–3666

Beaudet M, Brisson J, Gravel D, Messier C (2007) Effect of a major canopy disturbance on the coexistence of *Acer saccharum* and *Fagus grandifolia* in the understorey of an old-growth forest. J Ecol 95:458–467

Canham CD (1988) Growth and canopy architecture of shade-tolerant trees: response to canopy gaps. Ecology 69:786–795

Cleavitt NL, Fairbairn M, Fahey TJ (2008) Growth and survivorship of American beech (*Fagus grandifolia* Ehrh.) seedlings in a northern hardwood forest following a mast event. J Torrey Bot Soc 135:328–345

Dix RI, Skrentny RF Jr (1965) Reproduction of *Fagus grandifolia* Ehrh by seed in Wisconsin. Can J Bot 43:757–763

Fang J, Lechowicz MJ (2006) Climatic limits for the present distribution of beech (*Fagus* L.) species in the world. J Biogeogr 33:1804–1819

Forcier LK (1973) Seedling pattern and population dynamics, and the reproductive strategies of sugar maple, beech and yellow birch at Hubbard Brook. Ph.D. thesis, Yale University, New Haven

Gaskin GJ, Miller JD (1996) Measurement of soil water content using a simplified impedance measuring technique. J Agric Res 63:153–160

Held ME (1983) Patterns of beech regeneration in the east-central United States. Torrey Bot Club 110(1):55–62

Houston DR (2001) Effect of harvesting regime on beech root sprouts and seedlings in a North-Central Maine forest long affected by beech bark disease. Research Paper NE-717. USDA Forest Service, Newtown Square, PA, USA

Jones RH, Raynal DJ (1987) Root sprouting in American beech: production, survival, and the effect of parent tree vigor. Can J For Res 17:539–544

Jones RH, Raynal DJ (1988) Root sprouting in American beech (*Fagus grandifolia*): effects of root injury, root exposure, and season. For Ecol Manag 25:79–90

Kitamura K, Kawano S (2001) Regional differentiation in genetic components for the American beech, *Fagus grandifolia* Ehrh., in relation to geological history and mode of reproduction. J Plant Res 114:353–368

Kochenderfer JD, Kochenderfer JN, Warner A, Miller GM (2004) Preharvest manual herbicide treatments for controlling American beech in central West Virginia. North J Appl For 21:40–49

Kochenderfer JD, Kochenderfer JN, Miller GM (2006) Controlling beech root and stump sprouts using the cut-stump method. North J Appl For 23:155–165

Morris AB, Small RL, Cruzan MB (2004) Variation in frequency of clonal reproduction among populations of *Fagus grandifolia* Ehrh. in response to disturbance. Castanea 69:38–51

Niinemets U (2001) Global-scale climatic controls of leaf dry mass per area, density, and thickness in trees and shrubs. Ecology 82:453–469

Nyland RD (2008) Origin of small understory beech in New York northern hardwood stands. North J Appl For 25:161–163

South DB (1995) Relative growth rates: a critique. S Afr For J 173:43–48

Takahashi K, Lechowicz MJ (2008) Do interspecific differences in sapling growth traits contribute to the co-dominance of *Acer saccharum* and *Fagus grandifolia*? Ann Bot 101:103–109

Takahashi K, Arii K, Lechowicz MJ (2010) Codominance of *Acer saccharum* and *Fagus grandifolia*: the role of *Fagus* root sprouts along a slope gradient in an old-growth forest. J Plant Res 123:665–674

Tubbs CH, Houston DR (1990) *Fagus grandifolia* Ehrh. In: Burns RM, Honkala BH (eds) Silvics of North America: 2. Hardwoods, Agriculture handbook 654. US Department of Agriculture, forest service, Washington, pp 325–332

Uemura A, Ishida A, Matsumoto Y (2005) Simulated seasonal changes of CO_2 and H_2O exchange at the top canopies of two *Fagus* trees in a winter-deciduous forest, Japan. For Ecol Manag 212:230–242

Wagner S, Collet C, Madsen P, Nakashizuka T, Nyland RD, Sagheb-Talebig K (2010) Beech regeneration research: from ecological to silvicultural aspects. For Ecol Manag 259:2172–2182

Ward RT (1961) Some aspects of regeneration habits of the American beech. Ecology 42:828–832

Wiehle M, Eusemann P, Thevs N, Schnittler M (2009) Root suckering patterns in *Populus euphratica* (Euphrates poplar, Salicaceae). Trees Struct Funct 23:991–1001